

Introduction

Radox provides an API Integration Suite to allowing connectivity to the following systems:

- Nexus Lab Test Order Portal
- Home Test Kit (HTK) Dispatch Portal
- Clinic Appointment Booking Portal

Nexus Lab Test Order Portal

This is the core portal within the integration suite which forms the backbone of all integration to the Radox systems. This allows both Radox and the Corporate Customer to track the status of any test order from initial creation through to results delivery providing multiple status updates through the test sample's journey. Unless an alternative flow is agreed with Radox this should be the starting point for any integration.

Home Test Kit (HTK) Dispatch Portal

If a Home Test Kit forms part of the agreed business flow then this portal and its underlying API suite will allow the test order to be linked to the dispatch of a home sample collection kit. This will be dispatched to the provided address of the customer/patient. Internal automated processes will update the Nexus test order record which can be monitoring by the corporate customer.

Clinic Appointment Booking Portal

If a Clinic Appointment for sample collection at a clinic within the Radox network is part of the agreed business flow then this portal and its underlying API suite will allow the test order to be linked to a clinic appointment. You will be able to create a booking flow for this customer / patient to select a location, appointment date and appointment time. Again, internal automated processes will update the Nexus test order record which can be monitoring by the corporate customer.

High Level Data Flow

Below is a high-level data / API flow diagram for connectivity to the Radox API Integration Suite. The following are the proposed steps in this flow. Please note the actual integration to the API suites will be more complex than this high-level flow.

Detailed API documentation can be found at <https://developers.radox.com/>.

Step 1 – Placing Order

It is envisaged that the end customer / patient will be interacting with the corporate customer's own corporate systems in a potential membership or e-commerce style site. Product offerings, already pre-agreed with Radox, will be on display for the customer / patient to review, interact with and purchase.

Although product identifiers will be required for the ultimate creation of test orders and their underlying sample collection method (home or in clinic) these identifiers will usually be agreed with the corporate customer in advance and stored within the corporate customer's own systems. The exception to this will be the in-clinic appointment flow where corporate systems will need to utilise the APIs within the Clinic Booking Portal for a live view of locations and underlying appointment dates/times.

Primary endpoints for the in-clinic flow during this period are:

- GetServiceAndAddons – this will most likely be pre-agreed with Radox and necessary details already stored within the Corporate Customer's systems.

- **GetServiceLocations** – this will provide a list of clinic details including latitude, longitude and postal address for the chosen service. Latitude and longitude coordinates can be utilized by the corporate customer's system to provide a "closest to" list depending on the end customer / patient's town or postal code (note this must be calculated by the corporate customer's own systems).
- **AvailabilityDetails** – based on the chosen service, location and Search From date this endpoint will present a list of appointment dates and time slots. The number of days presented is controlled by Randex. This is usually 7 days of available appointments slots for primary clinics and a longer period for pop-up style locations. The objective is to present 7 dates of available appointments, which depending on the availability, may not be consecutive dates.

Step 2 – Nexus Test Order Creation

Key to the entire process will be the creation of a Nexus Test Order record which acts as the central data record for on-going monitoring.

Primary endpoints are:

- **CreatePendingOrder** – the creation of the overall test order within the Nexus Lab portal. This contains details about the end customer / patient and all the panels and/or individual tests that are to be tested once sample collection has been completed. This endpoint will return the two test order identifiers which must be utilized throughout the entire process.
- **UpdatePendingOrder** – there is a window of opportunity (depending on the sample collection method) for the test order to be updated.
- **CancelOrder** – there is a window of opportunity (depending on the sample collection method) for the test order to be cancelled. The original test order will be retained within the portal but moved to a cancelled status.
- **GetOrderStatus** – this end point should be used as the central monitoring method to track the sample through the entire process. In due course monitoring options will be enhanced with the addition of appropriate web hooks but at this time this polling endpoint must be used. To spread the load on all systems a suitable polling strategy should be implemented. The suggested strategy would be polling each outstanding test order once per hour based on the test order creation date/time.

Step 3 – Sample Collection Method Decision

Depending on the complexity and quantity of the panels being offered to the end customer / patient there is a possible choice of two sample collection methods (Home Sample Collection and In-clinic Sample Collection). The rules around which choices can be offered to the end customer / patient will depend on the contractual agreements put in place with the Randex Business Team and fall outside of the scope of this document.

Step 4 – Sample Collection Method Creation

In the step the Home Dispatch or Clinic Booking records will be placed.

Home Dispatch

Primary endpoints are:

- **CreateOrder** – the creation of the dispatch request for the home sample collection kit. This contains details about the Nexus test order number, end customer / patient name, address delivery details, contact information and the kit code to be dispatched. This process will result in direct communication from Royal Mail to the end customer on the delivery status. Automated internal processes update the test order within Nexus where the status will move to status 2 (confirmed) once the package is ready for dispatch.
- **GetOrderStatusDetails** – once the Nexus status moves to status 2 then this end point can be used to get the dispatch tracking information in addition to the kit URNs (unique reference numbers) for each dispatched kit.

Clinic Booking

Primary endpoints are:

- HoldAvailabilityBooking – you can temporarily hold an appointment slot to allow time for the end customer / patient to complete the booking which might include payment for example. Slots will be held for a 30 minute period.
- CreateRandoxBooking – allows the creation of the permanent clinic booking record. This end point requires all information about the end customer, appointment information and service being booked. The Nexus test order number should also be provided as the link to the test order record.
- CancelRandoxBooking – there is a window of opportunity for the clinic booking record to be cancelled. If the entire test order is being cancelled then the Nexus test order record must also be cancelled separately.
- RescheduleAppointment – there is a window of opportunity for the clinic booking record to be rescheduled to a different clinic location, date and time.

Step 5 – Sample Collection by Customer / Patient

This is a physical step in the process where the end customer / patient will have an appropriate sample collection taken. Although outside the technical scope this can in theory occur as follows:

- Home Sample Collection Kit – a kit will be delivered to the customer's provided address. This kit can be either:
 - Standard Finger Prick sample collection carried out at home.
 - Premium Tasso sample collection also carried out at home.
 - Venepuncture kit which can be taken independently by the customer / patient to an independent phlebotomy service.
- In-clinic Sample Collection at a clinic within the Randox Clinic Network.

Primary endpoints during this period are:

- GetOrderStatus – continued monitoring can be achieved through the polling of this endpoint.

Step 6 – Sample Sent to the Laboratory for Testing

The physical sample arrives at the Randox Laboratory either via pre-paid postage in the case of a Home Sample Collection Kit posted by the end customer / patient or via the Randox Clinic in the case of in-clinic sample collection. The sample will be accessioned by the Randox Laboratory Team and automated process will update the Nexus test order to a status of 3 (awaiting results).

Primary endpoints during this period are:

- GetOrderStatus – continued monitoring can be achieved through the polling of this endpoint.

Step 7 – Results Now Available

Once the laboratory process has been completed and results approved for release automated process will update the Nexus test order to a status of 4 (complete). Please note that results can also contain various void codes and result caveat codes. Details of these codes can be provided by the Randox Business Team. In the event that all results have been voided then the status will automatically move to status 5 (cancelled).

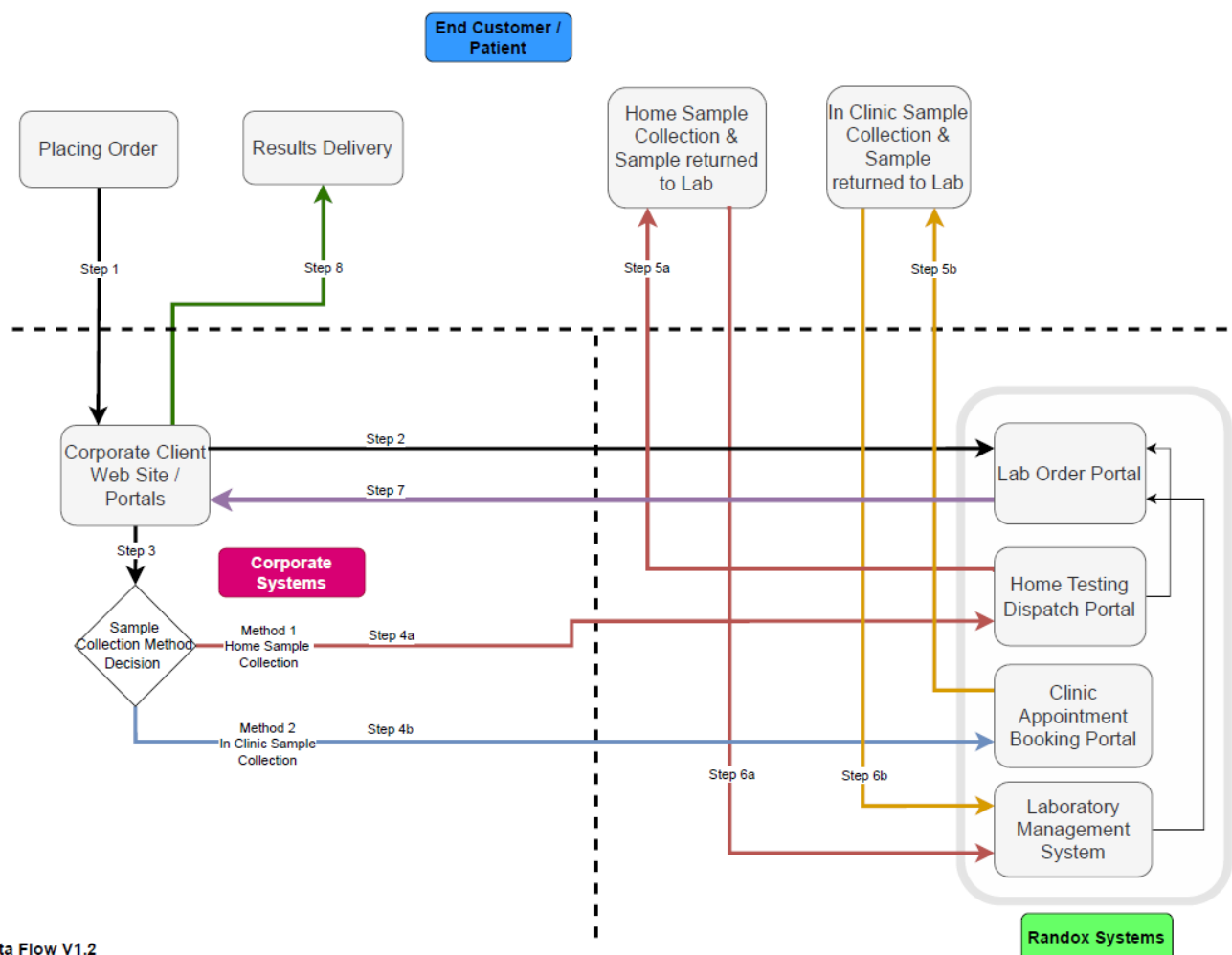
Primary endpoints during this period are:

- GetOrderStatus – continued monitoring can be achieved through the polling of this endpoint.

- GetOrderResultDetail – this endpoint provides a detailed JSON object containing the results value and any associated data (reference ranges, high/low indicators and any caveat codes) for each test carried out.
- GetOrderResultReports – this endpoint provides a pre-formatted results PDF file which can be used internally or presented to the end customer / patient. If an end customer-facing report has been requested and successfully generated, then this will be presented. If this report was not requested or cannot be generated, then a laboratory tabular report will be provided.

Step 8 – Results Provided to Customer / Patient

Detailed results can now be presented to the end customer / patient in whatever format / process is required by the corporate customer.



API Data Flow V1.2